
LECTURE NOTES

Lecture 8: Beyond Local Optima

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Benchmarking

Stochastic Methods

- Randomness can help escape local minima
- Control over randomness and the exploration vs exploitation trade off.

Noisy descent

Saddle points, where the gradient is very close to zero, can cause descent methods to select step sizes that are too small to be useful

We add Gaussian noise at each descent step

$$x_{k+1} = x_k + \alpha \nabla f(x_k) + \varepsilon_k, \quad \varepsilon_k \sim \mathcal{N}(0, \sigma_k^2)$$

- $\sigma_k = \frac{1}{k}$

Stochastic Gradient Descent

We can approximately evaluate gradients using randomly chosen subsets of the training data (batches)

This is much less computationally expensive than calculating the true gradient at every iteration. It yields approximately the same effect as noisy gradient approximation.

This ensures that the positive step sizes be chosen such that:

$$\sum_{k=1}^{\infty} a_k = \infty, \quad \sum_{k=1}^{\infty} \alpha_k^2 < \infty$$

Allows the step sizes to decrease and allow the method to converge, but not too quickly so as to become stuck away from a local minimum

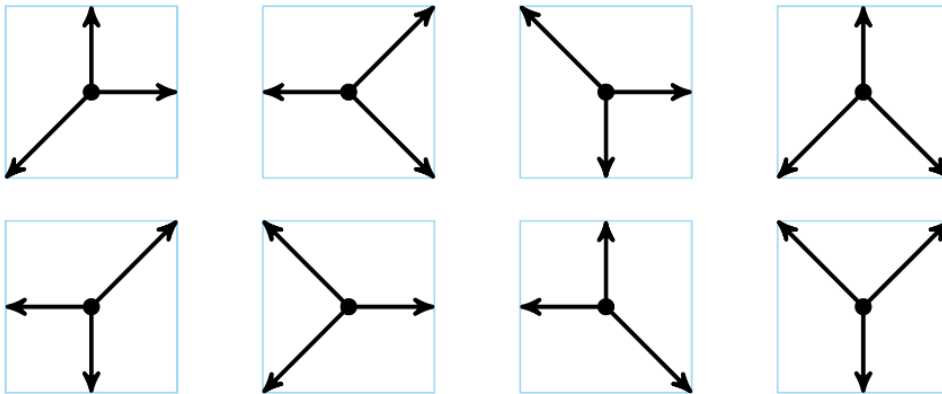


Figure 1:

Mesh Adaptive Direct Search

Similar to generalized pattern search but uses random positive spanning directions

Example: set of positive spanning sets constructed from nonzero directions $d_1, d_2 \in \{-1, 0, 1\}$.

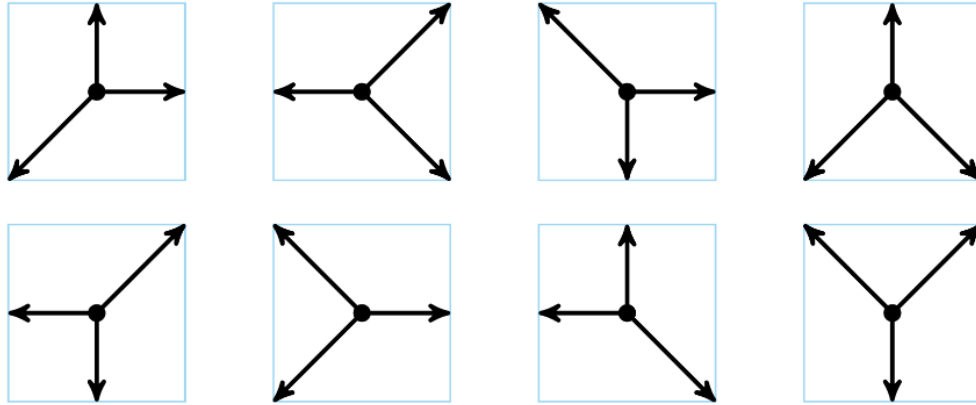


Figure 2:

Simulated Annealing

often used on functions with many local minima due to its ability to escape local minima.

A candidate transition from x to x' is sampled from a transition distribution T , eg, multivariate Gaussian

$$x' = x + \varepsilon, \quad \varepsilon \sim T$$

Metropolis acceptance criterion:

Simulated annealing

$$p(x, x') = \begin{cases} 1 & \text{if } \delta \leq 0 \\ e^{-\frac{\delta}{t_k}} & \text{if } \delta > 0 \end{cases}$$

Where $\delta = f(x') - f(x)$ is the change in the change in the objective value and t_k is the **temperature** at iteration k .

We can also do variable step-size v (separate direction components) in order to not use covariance matrix as a hyperparameter.

A cycle of random moves, one in each direction

$$x' = x + rv_i e_i$$

where r is randomly sampled from $\{-1, 1\}$

Then after n_s cycles, adjust the step size

$$\begin{cases} v_i \left(1 + c_i \frac{a_i - 0.6}{0.4} \right) & \text{if } a_i > 0.6n_s \\ v_i^- \left(1 + c_i \frac{0.4 - a_i}{0.4} \right) & \text{if } a_i < 0.4n_s \\ v_i & \text{if otherwise} \end{cases}$$

Annealing plan

t_k can decrease in different ways

- logarithmic annealing schedule

$$t_{k+1} = t_0 \cdot \frac{\ln(n)}{\ln(k+1)}$$

- exponential annealing schedule

$$t_{k+1} = \gamma t_k$$

- fast annealing

$$t_{k+1} = \frac{t_0}{k}$$

We can apply the decrease at different times.

Either at every proposition (either accepted or rejected), every move or at every m_t iteration.